Document Title

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1 Introduction

1.1 Matérn covariance function and its parametrizations

From [2], the Matérn covariance function is defined as

$$r(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{2\nu}h}{\rho} \right)^{\nu} K_{\nu} \left(\frac{\sqrt{2\nu}h}{\rho} \right) \tag{1}$$

and its corresponding spectral density is

$$\hat{f}(\omega) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + 4\pi^2\omega^2 \right)^{-(\nu+1/2)} \tag{2}$$

computed by using the Fourier transform pair parametrization

$$\hat{f}(\omega) = \int_{\mathbb{R}} e^{-i2\pi\omega h} r(h) dh \iff r(h) = \int_{\mathbb{R}} e^{i2\pi\omega h} \hat{f}(\omega) d\omega \tag{3}$$

But actually we want the pair parametrization

$$\tilde{f}(\omega) = \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega h} \tilde{f}(\omega) d\omega \tag{4}$$

For that, we simply do

$$\tilde{f}(\omega) = \hat{f}\left(\frac{\omega}{2\pi} \right) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{5}$$

But actually we apparently want this Fourier pair parametrization

$$\overline{f}(\omega) = \mathfrak{F}[r(h)](\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \mathfrak{F}^{-1}[\overline{f}(\omega)](h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} \overline{f}(\omega) d\omega \tag{6}$$

For that, we simply do

$$\overline{f}(\omega) = \frac{1}{\sqrt{2\pi}} \tilde{f}(\omega) = \frac{\sigma^2 \sqrt{2}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{7}$$

Recall the scaling property of the Fourier transform given by

$$r(ah) \iff \frac{1}{|a|} \overline{f}\left(\frac{\omega}{a} \right) \tag{8}$$

Now let $\varrho(h) = r(2h)$ and $\kappa = \sqrt{8\nu}/\rho$, then

$$\varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{8\nu}h}{\rho} \right)^{\nu} K_\nu \left(\frac{\sqrt{8\nu}h}{\rho} \right) \tag{9}$$

$$= \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{10}$$

which is the familiar form of the Matérn covariance function. Just for the record, if we have it in terms of τ, then

$$\varrho(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)} \frac{\Gamma(\nu)}{\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{11}$$

$$= \frac{1}{2^{\nu-1}\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{12}$$

Setting $\nu = \frac{1}{2}$, we obtain

$$\varrho(h) = \frac{1}{2^{-1/2}\Gamma(1)\sqrt{4\pi}\kappa^{1}\tau^2} (\kappa h)^{1/2} K_{1/2}(\kappa h). \tag{13}$$

Using the identities

$$\Gamma(1) = 1, \tag{14}$$

$$K_{1/2}(x) = \sqrt{\frac{\pi}{2x}} e^{-x}, \tag{15}$$

we get

$$(\kappa h)^{1/2} K_{1/2}(\kappa h) = (\kappa h)^{1/2} \sqrt{\frac{\pi}{2\kappa h}} e^{-\kappa h} = \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{16}$$

Substituting back,

$$\varrho(h) = \frac{1}{2^{-1/2}\sqrt{4\pi}\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{17}$$

Now simplify the constants:

$$2^{-1/2}\sqrt{4\pi} = \frac{1}{\sqrt{2}} \cdot 2\sqrt{\pi} = \sqrt{2\pi}, \tag{18}$$

so that

$$\varrho(h) = \frac{1}{\sqrt{2\pi}\,\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h} = \frac{1}{2\kappa\tau^2} e^{-\kappa h}. \tag{19}$$

By the scaling property, the corresponding spectral density to $\varrho(h)$ is $f(\omega) = \frac{1}{2}\overline{f}\left(\frac{\omega}{2}\right)$ given by

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \frac{\omega^2}{4} \right)^{-(\nu+1/2)} \tag{20}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{1}{4} \right)^{-(\nu+1/2)} \left(\frac{8\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{21}$$

Since $\left(\frac{1}{4}\right)^{-(\nu+1/2)} = 2 \cdot 4^{\nu}$ and $\kappa^2 = 8\nu/\rho^2$ we have that

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)}{\Gamma(\nu)}\left(\frac{8\nu}{\rho^2}\right)^{\nu}\left(\frac{8\nu}{\rho^2}+\omega^2\right)^{-(\nu+1/2)} \tag{22}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \tag{23}$$

Let $A = A_{(\nu,\kappa)} = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}$ so this matches with [1]. If $\sigma^2 = \dfrac{\Gamma(\nu)}{\Gamma(\nu+1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2}$, then $A\sigma^2 = (2\pi)^{-1/2}\tau^{-2}$. The moral of the story is that

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{24}$$

Putting in terms of α, we instead have

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-\alpha} \iff \varrho(h) = \frac{\sigma^2}{2^{\alpha-1/2-1}\Gamma(\alpha-1/2)}(\kappa h)^{\alpha-1/2}K_{\alpha-1/2}(\kappa h), \tag{25}$$

If we drop σ^2, we still have the Fourier pair

$$f_\star(\omega) = A\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho_\star(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{26}$$

1.2 Computational quirks

The `MetricGraph` package actually codifies

$$\overline{\varrho}(h;\kappa,\nu,\texttt{cte}) = \frac{\texttt{cte}^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{27}$$

So if you want something like

$$\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa,\nu,\sigma) = \frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{28}$$

computationally you should actually do

$$\sigma^2\overline{\varrho}\left(h;\kappa,\nu,\sqrt{\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}}\right) \tag{29}$$

1.3 Preliminaries

We consider the Matérn covariance function

$$\varrho(h;\kappa,\nu,\sigma) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|), \quad \sigma^2 = A^{-1}(2\pi)^{-1/2}\tau^{-2}, \tag{30}$$

where $\rho = \sqrt{8\nu}/\kappa$ and $\nu = \alpha - 1/2$ and the SPDE

$$(\kappa^2 - \Delta_\Gamma)^{\alpha/2}(\tau u) = \mathcal{W}, \qquad \text{on } \Gamma, \tag{31}$$

1.4 Computing spectral density from SPDE

We take the Fourier transform (6) to (31) to get

$$(\kappa^2+\omega^2)^{\alpha/2}\tau\hat{u}(\omega) = \widehat{\mathcal{W}}(\omega) \implies f(\omega) = \frac{1}{\tau^2\sqrt{2\pi}}(\kappa^2+\omega^2)^{-\alpha} \tag{32}$$

Let $L = (\kappa^2 - \Delta_\Gamma)$. The solution u is a centered Gaussian random field with covariance operator $\mathcal{C} = \tau^{-2}L^{-\alpha}$. We consider the approximation

$$\mathcal{C}_\alpha = \tau^{-2}L^{-\lfloor\alpha\rfloor}P_m(L)Q_m(L)^{-1} \tag{33}$$

where

$$x^{\{\alpha\}} \approx \frac{p_m(x)}{q_m(x)} = \frac{\sum_{i=0}^m a_i x^i}{\sum_{j=0}^m b_j x^j} = \frac{P_m(x^{-1})}{Q_m(x^{-1})} = r_m(x) \tag{34}$$

or

$$x^{-\{\alpha\}} \approx \frac{p_m(x^{-1})}{q_m(x^{-1})} = \frac{\sum_{i=0}^m a_i x^{m-i}}{\sum_{j=0}^m b_j x^{m-j}} = \frac{P_m(x)}{Q_m(x)} = R_m(x) \tag{35}$$

We further have that

$$\frac{P_m(x)}{Q_m(x)} = \frac{p_m(x^{-1})}{q_m(x^{-1})} = k + \sum_{i=1}^m r_i(x-p_i)^{-1}. \tag{36}$$

Therefore

$$\mathcal{C}_\alpha = \tau^{-2}L^{-\lfloor\alpha\rfloor}\left(kI + \sum_{i=1}^m r_i(L-p_iI)^{-1}\right) = \tau^{-2}kL^{-\lfloor\alpha\rfloor} + \sum_{i=1}^m \tau^{-2}r_iL^{-\lfloor\alpha\rfloor}(L-p_iI)^{-1} \tag{37}$$

Since $k, r_i > 0$ and $p_i < 0$, $\mathcal{C}_0 = \tau^{-2} k L^{-\lfloor \alpha \rfloor}$ and $\mathcal{C}_i = \tau^{-2} r_i L^{-\lfloor \alpha \rfloor}(L - p_i I)^{-1}$ are valid covariance operators so that $\mathcal{C}_\alpha = \mathcal{C}_0 + \sum_{i=1}^m \mathcal{C}_i$. Then the approximation u_α has decomposition

$$u_\alpha = u_0 + \sum_{i=1}^m u_i \tag{38}$$

where $u_0 \sim N(0, \mathcal{C}_0)$ and $u_i \sim N(0, \mathcal{C}_i)$. Moreover, they satisfy

$$\mathcal{C}_0^{-1/2} u_0 = L^{\lfloor \alpha \rfloor / 2}(\tau k^{-1/2} u_0) = \mathcal{W}, \qquad \text{on } \Gamma, \tag{39}$$

and

$$\mathcal{C}_i^{-1/2} u_i = (L^{\lfloor \alpha \rfloor}(L - p_i I))^{1/2}(\tau r_i^{-1/2} u_i) = \mathcal{W}, \qquad \text{on } \Gamma \tag{40}$$

Definition 1.1. u_i in (40) is called a shifted Whittle-Matérn process on Γ.

Remark 1. Observe that (39) are Whittle-Matérn process where the power $\lfloor \alpha \rfloor$ is integer.

Observe now that (30) has spectral density

$$f(\omega) = A\sigma^2(\kappa^2 + \omega^2)^{-(\nu + 1/2)}, \quad A = \sigma^{-2}(2\pi)^{-1/2}\tau^{-2} \tag{41}$$

which can be also written as

$$f(\omega) = A\sigma^2\kappa^{-2\alpha}(1 + \kappa^{-2}\omega^2)^{-\alpha} = A\sigma^2\kappa^{-2\alpha}(1 + \kappa^{-2}\omega^2)^{-\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2)^{-\{\alpha\}} \tag{42}$$

By (35), we can approximate f by f_α given by

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}}\frac{p_m(1 + \kappa^{-2}\omega^2)}{q_m(1 + \kappa^{-2}\omega^2)} \tag{43}$$

By (36), we have that

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}}\left(k + \sum_{i=1}^m r_i(1 + \kappa^{-2}\omega^2 - p_i)^{-1}\right) \tag{44}$$

$$= \frac{A\sigma^2\kappa^{-2\alpha}k}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}} + \sum_{i=1}^m \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2 - p_i)} \tag{45}$$

Let

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2 - p_i)} \tag{46}$$

then we can write

$$f_\alpha(\omega) = f_0(\omega) + \sum_{i=1}^m f_i(\omega), \tag{47}$$

which is a valid decomposition of spectral densities. We now can equivalently write

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k\kappa^{2\lfloor \alpha \rfloor}}{(\kappa^2 + \omega^2)^{\lfloor \alpha \rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i\kappa^{2\lfloor \alpha \rfloor + 2}}{(\kappa^2 + \omega^2)^{\lfloor \alpha \rfloor}(\kappa^2(1 - p_i) + \omega^2)} \tag{48}$$

Now for $k \in \mathbb{N}$ and $p \in \mathbb{R}$, we have that

$$\frac{1}{(\kappa^2 + \omega^2)^k(\kappa^2(1 - p) + \omega^2)} = \frac{1}{(\kappa^2 p)^k(\kappa^2(1 - p) + \omega^2)} - \sum_{j=1}^k \frac{1}{(\kappa^2 p)^{k+1-j}(\kappa^2 + \omega^2)^j} \tag{49}$$

Indeed, let $r = \dfrac{\kappa^2 p}{\kappa^2+\omega^2}$, then $1-r = \dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}$. Hence

$$\begin{aligned}
\sum_{j=1}^{k} \frac{1}{(\kappa^2 p)^{k+1-j}(\kappa^2+\omega^2)^j} &= \frac{1}{(\kappa^2 p)^{k+1}} \sum_{j=1}^{k} \left(\frac{\kappa^2 p}{\kappa^2+\omega^2}\right)^j \\
&= \frac{1}{(\kappa^2 p)^{k+1}} \frac{r(1-r^k)}{1-r} \\
&= \frac{1}{(\kappa^2 p)^{k+1}} \frac{\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\left(1-\left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^k\right)}{\dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}} \\
&= \frac{1-\left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^k}{(\kappa^2 p)^k(\kappa^2(1-p)+\omega^2)} \\
&= \frac{1}{(\kappa^2 p)^k(\kappa^2(1-p)+\omega^2)} - \frac{1}{(\kappa^2+\omega^2)^k(\kappa^2(1-p)+\omega^2)}
\end{aligned}$$

Therefore

$$f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}(\kappa^2(1-p_i)+\omega^2)} \tag{50}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A\sigma^2\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}(\kappa^2+\omega^2)^j} \tag{51}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}} \sigma^2(\kappa^2(1-p_i)+\omega^2)^{-1} \tag{52}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}} \sigma^2(\kappa^2+\omega^2)^{-j} \tag{53}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor} A_{(1/2,\kappa\sqrt{1-p_i})}} A_{(1/2,\kappa\sqrt{1-p_i})} \sigma^2(\kappa^2(1-p_i)+\omega^2)^{-(1/2+1/2)} \tag{54}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j} A_{(j-1/2,\kappa)}} A_{(j-1/2,\kappa)} \sigma^2(\kappa^2+\omega^2)^{-(j-1/2+1/2)} \tag{55}$$

$$= f_{i0}(\omega) - \sum_{j=1}^{\lfloor\alpha\rfloor} f_{ij}(\omega) \tag{56}$$

$$\begin{cases}
\varrho(h) & = \mathfrak{F}^{-1}[f(\omega)](h) = \dfrac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^\nu K_\nu(\kappa h) \\
f(\omega) & = A_{(\nu,\kappa)}\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \\
A_{(\nu,\kappa)} & = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)} = \dfrac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)} \\
\sigma^2 & = A_{(\nu,\kappa)}^{-1}(2\pi)^{-1/2}\tau^{-2} \\
\rho & = \sqrt{8\nu}/\kappa \\
\nu & = \alpha - 1/2
\end{cases} \tag{57}$$

Similarly for f_0

$$f_0(\omega) = A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}\sigma^2(\kappa^2+\omega^2)^{-\lfloor\alpha\rfloor} = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}A_{(\lfloor\alpha\rfloor-1/2,\kappa)}\sigma^2(\kappa^2+\omega^2)^{-(\lfloor\alpha\rfloor-1/2+1/2)} \tag{58}$$

By taking the inverse Fourier transform, we arrive at

$$\varrho_0(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,\sigma) \tag{59}$$

and

$$\varrho_i(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) - \sum_{j=1}^{\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,\sigma) \tag{60}$$

Let us simplify the expressions above. We first simplify ϱ_0.

$$\begin{aligned}
\varrho_0(h) &= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{\frac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}}{\frac{\sqrt{2}\Gamma(\lfloor\alpha\rfloor-1/2+1/2)\kappa^{2(\lfloor\alpha\rfloor-1/2)}}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\kappa^{2\nu}\kappa^{-2(\lfloor\alpha\rfloor-1/2)}\frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+\nu-\lfloor\alpha\rfloor+1/2)}\frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\varrho\left(h;\kappa,\lfloor\alpha\rfloor-1/2,\sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right)
\end{aligned}$$

where $c_a = \Gamma(a)/\Gamma(a-1/2)$. For simplicity, let

$$\varrho_{i0}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) \tag{61}$$

and

$$\varrho_{ij}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,\sigma) \tag{62}$$

Let us now simplify ϱ_{i0}.

$$\begin{aligned}
\varrho_{i0}(h) &= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}}{A_{(1/2,\kappa\sqrt{1-p_i})}} \varrho(h;\kappa\sqrt{1-p_i},1/2,1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor} \frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(1/2+1/2)(\kappa\sqrt{1-p_i})^{2\cdot 1/2}}{\Gamma(1/2)}} \varrho(h;\kappa\sqrt{1-p_i},1/2,1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor+\nu-1/2)} \frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(1/2+1/2)\sqrt{1-p_i}}{\Gamma(1/2)}} \varrho(h;\kappa\sqrt{1-p_i},1/2,1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}} \varrho(h;\kappa\sqrt{1-p_i},1/2,1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \varrho\left(h;\kappa\sqrt{1-p_i},1/2,\sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right)
\end{aligned}$$

since $\Gamma(1)=1$ and $\Gamma(1/2)=\sqrt{\pi}$. Now we simplify ϱ_{ij}.

$$\begin{aligned}
\varrho_{ij}(h) &= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{A_{(\nu,\kappa)}}{A_{(j-1/2,\kappa)}} \varrho(h;\kappa,j-1/2,1) \\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(j)\kappa^{2(j-1/2)}}{\Gamma(j-1/2)}} \varrho(h;\kappa,j-1/2,1) \\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j+\nu-j+1/2)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{c_\alpha}{c_j} \varrho(h;\kappa,j-1/2,1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \varrho\left(h;\kappa,j-1/2,\sqrt{c_\alpha/c_j}\right)
\end{aligned}$$

To sum up, we have arrived at

$$\varrho_\alpha(h) = \varrho_0(h) + \sum_{i=1}^{m} \varrho_i(h) \tag{63}$$

where

$$\varrho_0(h) = k\sigma^2 \varrho\left(h;\kappa,\lfloor\alpha\rfloor-1/2,\sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right) \tag{64}$$

and

$$\varrho_i(h) = r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \varrho\left(h;\kappa\sqrt{1-p_i},1/2,\sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right) - \sum_{j=1}^{\lfloor\alpha\rfloor} r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \varrho\left(h;\kappa,j-1/2,\sqrt{c_\alpha/c_j}\right) \tag{65}$$

2 Comparison to rat int paper

The calculation above is a detailed derivation of the partial-fraction identity and inverse Fourier transforms underlying Proposition 5 of [1], equation (9) on page 7 of `rat_int_paper.pdf`; its proof appears on pages 28–29. The final component formulas agree after translating the notation, but some conventions, qualifications, and inconsistencies must be separated. The comparison below refers to the supplied PDF, including its printed formulas, rather than to a potentially revised version of the paper.

2.1 Notation and agreement of the component formulas

The component covariance formulas coincide with Proposition 5 after identifying our residues r_i with the paper's c_i and translating the Matérn parametrization:

$$\varrho(h;\kappa,\nu,s) = \varrho^{\text{paper}}(h;\nu,\kappa,s^2). \tag{66}$$

Our last argument is a standard deviation, whereas the paper's is a variance; this explains the square roots in our final arguments. For later reference, let $q = \lfloor \alpha \rfloor$, $\kappa_i = \kappa\sqrt{1-p_i}$, and $M_{\nu,\lambda}(h) = \varrho(h;\lambda,\nu,1)$. For $q \geq 1$, the common formulas are

$$\varrho_0(h) = k\sigma^2 \frac{c_\alpha}{c_q} M_{q-1/2,\kappa}(h), \tag{67}$$

$$\varrho_i(h) = r_i \sigma^2 c_\alpha \left[\frac{\sqrt{\pi}}{p_i^q \sqrt{1-p_i}} M_{1/2,\kappa_i}(h) - \sum_{j=1}^{q} \frac{M_{j-1/2,\kappa}(h)}{p_i^{q+1-j} c_j} \right]. \tag{68}$$

2.2 Fourier normalization: an inconsistency in the supplied paper

Equation (4) in the paper is wrong: its factor $1/(2\pi)$ must be $1/\sqrt{2\pi}$ to agree with the printed $A = \sqrt{2}\, c_\alpha \kappa^{2\alpha-1}$ and Proposition 5. The corrected formula is

$$r_{m,\alpha}(h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} f_{m,\alpha}(\omega)\, d\omega. \tag{69}$$

This is exactly the convention used in (6); after this correction, there is no Fourier normalization difference between the two derivations.

2.3 The range $1/2 < \alpha < 1$ and the white-noise component

For $1/2 < \alpha < 1$, we have $q = \lfloor \alpha \rfloor = 0$. Hence the zeroth term of the spectral decomposition becomes

$$f_0(\omega) = \frac{A\sigma^2 \kappa^{-2\alpha} k}{(1+\kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}} = A\sigma^2 \kappa^{-2\alpha} k. \tag{70}$$

Under the symmetric Fourier convention, $\mathfrak{F}^{-1}[1](h) = \sqrt{2\pi}\, \delta_0(h)$, since $\int_{\mathbb{R}} e^{i\omega h}\, d\omega = 2\pi\delta_0(h)$. Therefore,

$$\varrho_0(h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} A\sigma^2 \kappa^{-2\alpha} k\, d\omega \tag{71}$$

$$= \sqrt{2\pi} A\sigma^2 \kappa^{-2\alpha} k\, \delta_0(h). \tag{72}$$

Using $A = \sqrt{2}\, c_\alpha \kappa^{2\alpha-1}$ and $A\sigma^2 = (2\pi)^{-1/2}\tau^{-2}$ gives

$$\varrho_0(h) = b\, \delta_0(h), \qquad b = \sqrt{2\pi} A\sigma^2 \kappa^{-2\alpha} k = \frac{k\sigma^2 c_\alpha \sqrt{4\pi}}{\kappa} = \frac{k}{\tau^2 \kappa^{2\alpha}}. \tag{73}$$

2.4 Operator definition and scaling in the present derivation

There is an inconsistency in the operator paragraph preceding (35): defining $L = (\kappa^2 - \Delta_\Gamma)^{\alpha/2}$ and then writing $\mathcal{C} = \tau^{-2} L^{-\alpha}$ applies the exponent twice. Instead define the positive self-adjoint base operator, with its graph domain and vertex conditions specified, by

$$L_0 = \kappa^2 - \Delta_\Gamma, \qquad \mathcal{C} = \tau^{-2} L_0^{-\alpha}. \tag{74}$$

Equivalently, retaining the original powered L would require $\mathcal{C} = \tau^{-2} L^{-2}$. The paper's Section 5 uses an unpowered base operator in $L^\beta(\tau u) = \mathcal{W}$; the correspondence here is $L = L_0$ and $\beta = \alpha/2$. The covariance of u includes τ^{-2} (the paper's displayed $L^{-2\beta}$ suppresses this scale).

A second distinction is that equation (8) and our spectral calculation apply the rational approximation to the dimensionless variable $1 + \kappa^{-2}\omega^2$. To use the same coefficients k, r_i, p_i at operator level, define

$$B = \kappa^{-2}L_0 = I - \kappa^{-2}\Delta_\Gamma, \qquad R_m(x) = k + \sum_{i=1}^{m} \frac{r_i}{x - p_i}. \tag{75}$$

Then the consistent operator approximation and its components are

$$\mathcal{C}_m = \tau^{-2}\kappa^{-2\alpha}B^{-q}R_m(B) = \mathcal{C}_0 + \sum_{i=1}^{m} \mathcal{C}_i, \tag{76}$$

$$\mathcal{C}_0 = \tau^{-2}k\kappa^{2q-2\alpha}L_0^{-q}, \tag{77}$$

$$\mathcal{C}_i = \tau^{-2}r_i\kappa^{2q+2-2\alpha}L_0^{-q}(L_0 - \kappa^2 p_i I)^{-1}. \tag{78}$$

In particular, the shifted operator is $L_0 - \kappa^2 p_i I = \kappa_i^2 - \Delta_\Gamma$, consistent with the κ_i in Proposition 5. The unscaled operator expressions above (39) do not use these same dimensionless coefficients unless $\kappa = 1$ or the coefficients have been rescaled. In the unscaled variable the equivalent coefficients are

$$\widetilde{k} = k\kappa^{2q-2\alpha}, \qquad \widetilde{r}_i = r_i\kappa^{2q+2-2\alpha}, \qquad \widetilde{p}_i = \kappa^2 p_i. \tag{79}$$

For example, the component SPDEs consistent with (76) are

$$L_0^{q/2}\big(\tau k^{-1/2}\kappa^{\alpha-q}u_0\big) = \mathcal{W}_0, \tag{80}$$

$$\big[L_0^q(L_0 - \kappa^2 p_i I)\big]^{1/2}\big(\tau r_i^{-1/2}\kappa^{\alpha-q-1}u_i\big) = \mathcal{W}_i. \tag{81}$$

These scaling issues occur in the earlier operator formulation; the later spectral derivation already contains the required powers of κ and leads to the matching formulas (67)–(68).

2.5 Stationary interval covariances versus metric-graph covariances

Proposition 5 concerns stationary processes on $\mathbb{R}$ restricted to an interval, so their covariances depend on the lag $h = s - t$. An ordinary Fourier transform diagonalizes $\kappa^2 - \Delta$ on $\mathbb{R}$; it does not in general diagonalize a metric-graph Laplacian with vertex conditions. Consequently, the Fourier step applied to (31) must be understood as the whole-line calculation, not as a proof that a graph covariance has the same stationary lag formula.

For a compact metric graph with orthonormal eigenfunctions ϕ_n of $-\Delta_\Gamma$ and eigenvalues $\lambda_n \geq 0$, the corresponding covariance is instead

$$C_\Gamma(s,t) = \tau^{-2}\sum_n (\kappa^2 + \lambda_n)^{-\alpha}\phi_n(s)\phi_n(t), \tag{82}$$

with the rational approximation obtained by replacing each spectral multiplier according to (76). This generally depends on both s and t and includes topology and vertex conditions. Even on an interval, imposing Neumann conditions defines a different model from restricting a stationary whole-line Matérn process. The paper discusses operator extensions in Section 5, but Proposition 5 alone does not establish the graph construction or the required endpoint precision corrections and conditioning. Those are additional steps beyond the inverse Fourier calculation reproduced here.

2.6 Typographical issues in the paper's proof

The geometric-sum identity used here is the same one used on page 28. It requires $p \neq 0$; the earlier wording "$p \in \mathbb{R}$" should include that qualification, which is automatic for the actual poles

$p_i < 0$. Using q for the integer order also avoids confusing the identity's integer k with the partial-fraction constant k.

In the final displayed spectral decomposition on page 28 of the paper, the first resolvent term is printed inside the sum over j, and some denominators use p rather than p_i. Read literally, that display does not follow from equation (8). The consistent expression is

$$f_{m,\alpha}(\omega) = A\sigma^2\kappa^{-2\alpha}\left[\frac{k\kappa^{2q}}{(\kappa^2+\omega^2)^q} + \sum_{i=1}^{m} r_i\kappa^{2q+2}H_i(\omega)\right], \tag{83}$$

$$H_i(\omega) = \frac{1}{(\kappa^2 p_i)^q(\kappa_i^2+\omega^2)} - \sum_{j=1}^{q}\frac{1}{(\kappa^2 p_i)^{q+1-j}(\kappa^2+\omega^2)^j}. \tag{84}$$

Thus the first term occurs once for each i, outside the inner j-sum. The detailed derivation above already uses this placement and the indexed poles correctly. The discrepancy is in that printed proof display, not in the final component formulas of Proposition 5.

3 Implementation

3.1 Case $\alpha \in (0,1)$

$$\varrho_0(h) = k\sigma^2\frac{c_\alpha\sqrt{4\pi}}{\kappa}1_{[h=0]} \implies \mathbf{Q}_0 = \mathbf{I}\cdot\frac{1}{(k\sigma^2 c_\alpha\sqrt{4\pi})/\kappa} \tag{85}$$

$$\varrho_i(h) = r_i\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) \implies \mathbf{Q}_i = \frac{1}{(r_i c_\alpha\sqrt{\pi})/\sqrt{1-p_i}}\begin{cases}\overline{\nu} = 1/2\\ \overline{\kappa} = \kappa\sqrt{1-p_i}\\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}}\\ \cdot\texttt{Qalpha1}(\overline{\tau},\overline{\kappa})\end{cases} \tag{86}$$

3.2 Case $\alpha \in (1,2)$

$$\varrho_0(h) = k\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,\sigma) \implies \mathbf{Q}_0 = \frac{1}{kc_\alpha/c_{\lfloor\alpha\rfloor}}\begin{cases}\overline{\nu} = 1/2\\ \overline{\kappa} = \kappa\\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}}\\ \cdot\texttt{Qalpha1}(\overline{\tau},\overline{\kappa})\end{cases} \tag{87}$$

$$\varrho_i(h) = \varrho_i(h) \implies \mathbf{Q}_i = \frac{1}{r_i\sigma^2}\left(\texttt{matern.p.precision}(\kappa,p_i,\alpha,\ell_e) - f(\texttt{matern.p.joint}(\kappa,p_i,\alpha))\right) \tag{88}$$

References

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[2] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian processes for machine learning*. Adaptive Computation and Machine Learning. MIT Press, Cambridge, MA, 2006. https://doi.org/10.7551/mitpress/3206.001.0001.